

29/03/2020 Activity 1

1. # include <stack.h>

int main() {

int stack[10], top = -1;

int queue[10], front = 0, rear = -1;

int s, q;

stack operations:-

stack[++top] = s;

stack;

stack

[54]

push 54

[52]
[54]

push 52

[54]

pop()

remove 52

[55]
[54]

push 55

[62]
[55]
[54]

push (62)

s = pop() → remove 62

s = 62

Queue Operations:-

U

← rear
[21]
front

enqueue(21)

[24]
[21]

enqueue(24)

[24]

pop(21)

[28]
[24]

enqueue(28)

[31]
[28]
[24]

enqueue(31)

← dequeue
[31]
[28]
[24]
[21]

dequeue(31)

s = 62, q = 24

s + q = 62 + 24 = 86

2. postfix:- 10 5 + 60 6 / * 8 -

10 → [10]

5 → [10, 5]

+ → [15]

60 → [15, 60]

6 → [15, 60, 6]

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$1 \rightarrow [5, 10]$
 $* \rightarrow [150]$
 $8 \rightarrow [150, 8]$
 $- \rightarrow 142$

$150 - 8 = 142$

3 postfix :- 8 2 3 1 / 2 3 * + 5 1 * -

$8 \rightarrow [8]$
 $2 \rightarrow [8, 2]$
 $3 \rightarrow [8, 2, 3]$
 $1 \rightarrow [8, 8]$
 $/ \rightarrow [1]$
 $2 \rightarrow [1, 2]$
 $3 \rightarrow [1, 2, 3]$
 $* \rightarrow [1, 6]$

$2 \times 3 = 6$

$8 / 1 = 8$

$2 \times 3 = 6$

After the first * is evaluated stack



stack [1, 6]

deg/min
 24
 31
 28
 sum
 engine
 (32)

$a + b \times c - d \wedge e \wedge f$

input

stack

output

a

a

a

+

+

a

b

+

ab

*

+

abc

c

+

abc

-

-

abc * +

d

-

abc * + d

^

-

abc * + d

e

-

abc * + de

^

-

abc * + de

f

-

abc * + def

-

-

(abc * + def) -

end

	Operation	stack	popped
5	push [10]	[10]	
1	push [20]	[10, 20]	
	pop	[10]	20
	push [10]	[10, 10]	
	push [20]	[10, 10, 20]	
	pop	[10, 10]	20
	pop	[10]	10
	pop	[]	10
	push [20]	[20]	
	pop	[]	20

sequence of popped out: 20, 20, 10, 10, 20

Final
20/7

